

# Model for Calculating the Edelstein Effect in a Rashba Fermion System

## 1. Theoretical Framework and Hamiltonian

The system is modeled as a two-dimensional electron gas (2DEG) with Rashba spin-orbit coupling (RSOC). The Hamiltonian describing the Rashba fermion at the  $\Gamma$  point of the Brillouin zone is given by:

$$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma}) \quad (1)$$

where:

- $p$  is the momentum operator.
- $m$  is the effective carrier mass.
- $\alpha$  is the Rashba spin-orbit coupling strength.
- $\hat{z}$  is the unit vector perpendicular to the 2D plane.
- $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$  is the vector of Pauli matrices.

The energy dispersion relation for the two chiral (helicity) bands  $\nu = \pm$  is:

$$E_\nu(k) = \frac{\hbar^2 k^2}{2m} + \nu \alpha k \quad (6)$$

where  $k = |\mathbf{k}|$  and  $\nu = +1$  corresponds to the outer branch and  $\nu = -1$  to the inner branch (or vice versa depending on the sign convention of  $\alpha$ ). The helicity operator is defined as  $\hat{S} = \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})/p$ , with eigenvalues  $s = \pm 1$ .

## 2. Calculation of Magnetization (Edelstein Effect)

The Direct Edelstein Effect (DEE) describes the generation of a non-equilibrium spin density (magnetization)  $\mathbf{M}$  under an applied electric field  $\mathbf{E}$ . Within the semiclassical Boltzmann approach, the magnetization at first order in the electric field is given by:

$$\mathbf{M} = -\mu_b \sum_{\mathbf{k}, \nu} |e| (\mathbf{v}_\nu(\mathbf{k}) \cdot \mathbf{E}) \delta[E_\nu(\mathbf{k}) - E_F] \langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^\nu \quad (2)$$

where:

- $\mu_b$  is the Bohr magneton.
- $|e|$  is the elementary charge.
- $\mathbf{v}_\nu(\mathbf{k}) = \nabla_{\mathbf{k}} E_\nu(\mathbf{k})$  is the group velocity.
- $\bar{\tau}_{\mathbf{k}}^\nu$  is the transport lifetime (often assumed constant  $\tau$  for simplicity).
- $\langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^\nu$  is the spin expectation value for the eigenstates.

For the isotropic Rashba model, the spin expectation value is:

$$\langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^\pm = \frac{1}{k} \begin{pmatrix} \pm k_y \\ \mp k_x \\ 0 \end{pmatrix} = \begin{pmatrix} \pm \sin \theta \\ \mp \cos \theta \\ 0 \end{pmatrix} \quad (3)$$

where  $\theta$  is the angle between  $\mathbf{k}$  and the  $\hat{x}$  axis.

## 2.1. Magnetization Magnitude and Direction

Assuming an electric field applied along the  $\hat{x}$  direction,  $\mathbf{E} = E_x \hat{x}$ , the resulting magnetization is perpendicular to the electric field and lies in the plane (along  $\hat{y}$ ). The magnitude depends on the filling regime:

### High-Density Regime (HDR)

When both chiral bands are occupied ( $E_F > 0$  relative to the band crossing):

$$M_y = \frac{\mu_b |e| \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y \quad (8)$$

In this regime, the spin density is **constant** and independent of the Fermi energy  $E_F$ . It scales linearly with the spin-orbit coupling  $\alpha$  and the effective mass  $m$ .

### Low-Density Regime (LDR)

When only the lowest energy band is occupied ( $E_F < 0$  relative to the band crossing, or near the band minimum):

$$M_y = \frac{\mu_b |e| \tau}{2\pi} \sqrt{m^2 \alpha^2 + 2m E_F} [\hat{z} \times \mathbf{E}]_y \quad (9)$$

In this regime, the spin density increases linearly with the Fermi energy  $E_F$  for small  $E_F$ . For values of  $E_F$  around the band crossing, the expression can be expanded as:

$$M_y \approx \frac{\mu_b |e| \tau}{2\pi} \left( \alpha m + \frac{1}{2} \frac{E_F}{\alpha} \right) [\hat{z} \times \mathbf{E}]_y \quad (10)$$

## 2.2. Edelstein Susceptibility

The linear Edelstein susceptibility  $\chi_{ij}$  is defined by  $M_j = \chi_{ij} E_i$ . For the isotropic case with  $\mathbf{E} = E_x \hat{x}$ :

$$\chi_{xy} = -\chi_0 \sum_{\nu=\pm} \int d^2k \langle \sigma_y \rangle_{\mathbf{k}}^{\nu} \delta(E_{\mathbf{k}}^{\nu} - \mu) v_x^{\nu}(\mathbf{k}) \quad (7)$$

where  $\chi_0 = \frac{\tau |e| \mu_b S_{cell}}{4\pi^2 a}$ .

#### Parameter Dependencies:

- **Spin-Orbit Coupling ( $\alpha$ ):** In the HDR,  $\chi_{xy}$  increases linearly with  $\alpha$  (see Eq. 8).
- **Fermi Energy ( $E_F$ ):** In the HDR,  $\chi_{xy}$  is constant (plateau). In the LDR, it depends on  $\sqrt{m^2 \alpha^2 + 2m E_F}$ .
- **Chirality:** The contribution comes from the difference in transport times and Fermi momenta of the two chiral bands ( $\bar{\tau}^+ k_F^+ - \bar{\tau}^- k_F^-$ ).
- **Fermi Velocity:** Implicit in the group velocity  $\mathbf{v}_{\nu}(\mathbf{k})$  and the density of states.

### 3. Anisotropic Rashba Model ( $C_{2v}$ Symmetry)

If the system exhibits anisotropy in effective mass ( $m_x \neq m_y$ ) or Rashba parameters ( $\alpha_x \neq \alpha_y$ ), the Hamiltonian becomes:

$$\hat{H} = \frac{\hbar^2 k_x^2}{2m_x} + \frac{\hbar^2 k_y^2}{2m_y} + \alpha_y k_y \hat{\sigma}_x - \alpha_x k_x \hat{\sigma}_y \quad (11)$$

The Edelstein susceptibility in the HDR depends on the anisotropy ratios  $r_m = m_y/m_x$  and  $r_{\alpha} = \alpha_y/\alpha_x$ :

$$\frac{\chi_{xy}}{\chi_0}(r_m) = \frac{4\pi m_x \alpha r_m}{1 + \sqrt{r_m}} \quad (12a)$$

$$\frac{\chi_{xy}}{\chi_0}(r_{\alpha}) = \frac{4\pi m \alpha_x r_{\alpha}}{1 + r_{\alpha}} \quad (12b)$$

The susceptibility can be boosted by rendering  $r_m$  and  $r_{\alpha}$  greater than one.

### 4. Explicit Graphics Description

Based on the extracted data, the following graphics should be generated to visualize the model results:

#### 1. Edelstein Susceptibility vs. Chemical Potential ( $\mu$ ):

- **Description:** Plot  $\chi_{xy}/\chi_0$  on the y-axis and  $\mu$  on the x-axis.

- **Behavior:** Shows a plateau in the High-Density Regime (constant value) and a rising curve in the Low-Density Regime.
- **Source:** Figure 2 (Left panel) and Figure 3 (Left panel) in Source [1].

## 2. Edelstein Susceptibility vs. Rashba Parameter ( $\alpha$ ):

- **Description:** Plot  $\chi_{xy}/\chi_0$  on the y-axis and  $\alpha$  on the x-axis at fixed  $\mu$ .
- **Behavior:** Linear increase with  $\alpha$  in the high-density regime.
- **Source:** Figure 3 (Right panel) in Source [1].

## 3. Fermi Surface and Spin Texture:

- **Description:** 2D plot in  $k_x$ - $k_y$  space showing the Fermi circles (inner and outer) and the spin orientation vectors (tangential to the circles).
- **Behavior:** Shows spin-momentum locking.
- **Source:** Figure 1 and Figure 2 (Right panel) in Source [1].

## 4. Anisotropy Dependence:

- **Description:** Surface plot or contour plot of  $\chi_{xy}/\chi_0$  as a function of  $r_m$  and  $r_\alpha$ .
- **Behavior:** Susceptibility increases with both ratios, saturating for large  $r_\alpha$ .
- **Source:** Figure 6 in Source [1].

# 5. Scientific Citations

### • Rashba Hamiltonian and Basic Formalism:

- Equation (1) and Introduction: Gaillardoni, I., et al., "Edelstein Effect in Isotropic and Anisotropic Rashba Models", *arXiv:2503.20712v1* (2025) [1].
- Original Prediction: Edelstein, V. M., "Spin polarization of conduction electrons induced by electric current in two-dimensional asymmetric electron systems", *Solid State Communications* **73**, 233 (1990) [29].

### • Analytical Expressions (HDR/LDR):

- Equations (8), (9), (10): Gaillardoni et al. (2025) [1].
- Derivations: Appendix A in Gaillardoni et al. (2025) [1].

### • Anisotropic Case:

- Equations (11), (12): Gaillardoni et al. (2025) [1].
- Derivations: Appendix B in Gaillardoni et al. (2025) [1].

### • Boltzmann Transport Approach:

- Equation (2): Gaillardoni et al. (2025) [1].
- Reference: Inoue, J.-i., et al., "Diffuse transport and spin accumulation in a rashba two-dimensional electron gas", *Physical Review B* **67**, 033104 (2003) [19].

## References:

- [1] I. Gaiardoni, M. Trama, A. Maiellaro, C. Guarcello, F. Romeo, and R. Citro, "Edelstein Effect in Isotropic and Anisotropic Rashba Models", *arXiv:2503.20712v1* [cond-mat.mes-hall] (2025).
- [29] V. M. Edelstein, "Spin polarization of conduction electrons induced by electric current in two-dimensional asymmetric electron systems", *Solid State Communications* **73**, 233 (1990).
- [19] J.-i. Inoue, G. E. Bauer, and L. W. Molenkamp, "Diffuse transport and spin accumulation in a rashba two-dimensional electron gas", *Physical Review B* **67**, 033104 (2003).